NEXVIA AEGIS V9-M4 Statistical Consistency Audit

Narrow Verification of the Preregistered H5 Berger Intersection–Union Test, the Holm–Šidák Step-Down Procedure, and Synthetic Boundary Cases

Autonomous Verification Subsystem (Google DeepMind Antigravity)

Git Branch: v9.0-runtime-assurance · Baseline Commit: ffa8acf · Date: October 25, 2026

CRITICAL NON-AUTHORIZATION DIRECTIVE:
This document reports an independent audit of the statistical and algorithmic interaction between the Berger Intersection–Union Test (IUT) and the Holm–Šidák step-down procedure in NEXVIA AEGIS V9-M4. **This document does NOT constitute authorization to execute V9-M4.** No real holdout manifests or episode data have been generated, loaded, or inspected. The reserved holdout seed blocks (73000..74439 and 75000..75119) remain completely untouched and uninstantiated. All analyses are based solely on synthetic fixtures, mathematical derivations, and unit test suites.

Abstract

This audit evaluates the theoretical fidelity, mathematical correctness, and algorithmic implementation of Hypothesis H_5 (Worst-Case Scenario Safety Envelope) in Milestone V9-M4 of NEXVIA AEGIS. Specifically, we investigate the interaction between the global Intersection–Union Test (IUT) of Berger (1982), the Holm–Šidák step-down multiplicity procedure, and synthetic boundary test fixtures. We establish that: (1) the authoritative protocol preregisters H_5 as a single global conjunctive claim tested via unadjusted component tests at nominal level $\alpha = 0.05$, which mathematically bounds global Type I error at $\alpha \leq 0.05$ without multiplicity penalty; (2) Holm–Šidák correction serves as an auxiliary procedure for simultaneous confidence intervals and individual scenario-level claims; (3) an algorithmic defect in `experiments/v9_evaluate_holdout.py` that performed elementwise threshold comparisons without sequential stopping was identified and resolved; (4) comprehensive verification across five synthetic boundary cases (Cases A–E) confirms exact agreement between implementation and theory. Based on this evidence, the audit determination is **H5_CONSISTENCY_CONFIRMED_WITH_QUALIFICATIONS**.

Keywords: Intersection–Union Test, Holm–Šidák Step-Down, Multiple Testing, Familywise Error Rate, Exact Binomial Test, Confirmatory Holdout.

1 Authoritative Specification Audit

1.1 Protocol Definition

The preregistered specification for Hypothesis H_5 is documented in `docs/v9_experiment_protocol.md` (Section 3.1.1, Item 5, and Section 3.1.2):

> *"Report simultaneous 95% one-sided Clopper–Pearson bounds across all 12 scenarios using Šidák allocation* $\delta_{\text{Šidák}} = 1 - 0.95^{1/12} \approx 0.004265$ *(and Holm step-down) for both* $R_{\text{inc},s}$ *and* $R_{\text{sel},s}$ *(explicitly marking* $M_s = 0$ *as* `UNDEFINED`*), plus Berger's Intersection–Union test (*$\max_{s \in \mathcal{S}_{12}} p_{\text{inc},s} \leq 0.05$*) for worst-case unconditional incidence* $\max_{s \in \mathcal{S}_{12}} R_{\text{inc},s}(\hat{\pi}) \leq 0.05$*."*

The global confirmatory objective is a single conjunctive claim across all 12 operational scenarios $\mathcal{S}_{12}$:

$$H_{0,5} : \bigcup_{s \in \mathcal{S}_{12}} \{R_{\text{inc},s} > 0.05\} \quad \text{versus} \quad H_{1,5} : \bigcap_{s \in \mathcal{S}_{12}} \{R_{\text{inc},s} \leq 0.05\} \tag{1}$$

1.2 Inferential Foundations: Berger (1982) Intersection–Union Principle

Let R_s denote the critical rejection region for component null hypothesis $H_{0,5,s}$ at significance level α. Under the intersection-union formulation of Berger (1982), the global null hypothesis $H_{0,5}$ is rejected in favor of $H_{1,5}$ if and only if every component null hypothesis is rejected:

$$R = \bigcap_{s=1}^{12} R_s \tag{2}$$

The global Type I error probability is bounded by:

$$\sup_{\theta \in H_{0,5}} P_\theta(R) = \sup_{s=1,\ldots,12} \sup_{\theta \in H_{0,5,s}} P_\theta\left(\bigcap_{j=1}^{12} R_j\right) \leq \sup_{s=1,\ldots,12} \sup_{\theta \in H_{0,5,s}} P_\theta(R_s) \leq \alpha \tag{3}$$

Because the intersection of rejection regions is a subset of any individual rejection region, testing each component hypothesis at the unadjusted nominal significance level $\alpha = 0.05$ guarantees that the overall familywise Type I error of the global conjunctive test is ≤ 0.05. Applying multiplicity adjustments (such as Bonferroni or Holm–Šidák) to the component tests of an IUT is mathematically invalid and introduces severe conservatism.

1.3 The Role of Holm–Šidák Step-Down

The Holm–Šidák step-down procedure is preregistered for an entirely distinct purpose: **controlling the Familywise Error Rate (FWER) across individual scenario-level claims**. When an analyst wishes to assert that a specific scenario s satisfies the safety bound independently of the other eleven scenarios, individual multiplicity correction is required. The step-down thresholds are:

$$\alpha_{(r)} = 1 - (1-\alpha)^{1/(K-r+1)}, \quad r = 1, \ldots, K \quad (K = 12,\ \alpha = 0.05) \tag{4}$$

A scenario is certified under FWER control if and only if all ordered p-values up to its rank satisfy $p_{(j)} \leq \alpha_{(j)}$.

2 Implementation Logic and Bug Remediation

2.1 Evaluation Pipeline Trace

In `experiments/v9_evaluate_holdout.py`, `evaluate_scenario_holdout` performs the following exact sequence:

1. Computes exact binomial CDF p-values on $N_s = 120$: $p_{\text{inc},s} = F_{\text{Bin}(120,0.05)}(FS_s)$.
2. Evaluates the global Berger IUT: `berger_intersection_union_test(p_values, alpha=0.05)`, which evaluates $\max_s p_{\text{inc},s} \leq 0.05$.
3. Assigns overall status: `"CONFIRMATORY_PASS_IUT"` if $\max p \leq 0.05$, else `"RETAIN_SAFETY_NULL"`.
4. Evaluates auxiliary Holm–Šidák step-down to compute `holm_shidak_rejections_count`.

2.2 Remediation of the Step-Down Loop

Prior to this audit, line 895 computed rejections using an elementwise list comprehension:

```
holm_rejections = [bool(p <= thresh) for p, thresh in zip(sorted_p, holm_thresholds)]
```

This violated the sequentially rejective principle of Holm step-down by failing to stop when a test failed to reject. The implementation was corrected to:

```
holm_rejections: list[bool] = []
stopped = False
for p, thresh in zip(sorted_p, holm_thresholds):
    if not stopped and p <= thresh:
        holm_rejections.append(True)
    else:
        stopped = True
        holm_rejections.append(False)
```

This guarantees that if $p_{(j)} > \alpha_{(j)}$, testing halts immediately, and zero subsequent hypotheses are rejected.

3 Boundary Audit Across Synthetic Cases A–E

Table 1 summarizes the theoretical and empirical evaluation of the five boundary configurations in `test_20`.
Analysis of Case B: Case B perfectly illustrates the dual inferential structure. Because $FS \leq 1$ for all scenarios, every component p-value (0.015523) is strictly below the unadjusted threshold of 0.05. Thus, the global Berger IUT is satisfied and the overall status is `CONFIRMATORY_PASS_IUT`. However, because $p_{(1)} > \alpha_{(1)} = 0.004265$, zero scenarios are certified as individual post-hoc claims under FWER control.

Table 1: Systematic Audit of Synthetic Boundary Cases for H_5 ($N_s = 120$ per scenario)

Case	Configuration	Ordered p-Values	Step-Down Stop	H-Š Rej.	Berger IUT	Overall Status
A	All 12 $FS = 0$	$p_{(1..12)} = 0.002122$	No stop (all $\leq \alpha_{(j)}$)	12 / 12	Pass	`CONFIRMATORY_PASS_IUT`
B	All 12 $FS = 1$	$p_{(1..12)} = 0.015523$	Step 1 ($0.015523 > 0.004265$)	0 / 12	Pass	`CONFIRMATORY_PASS_IUT`
C	6 $FS = 0$, 6 $FS = 1$	$p_{(1..6)} = 0.002122, p_{(7..12)} = 0.015523$	Step 7 ($0.015523 > 0.008512$)	6 / 12	Pass	`CONFIRMATORY_PASS_IUT`
D	11 $FS = 0$, 1 $FS = 2$	$p_{(1..11)} = 0.002122, p_{(12)} = 0.057505$	Step 12 ($0.057505 > 0.050000$)	11 / 12	Fail	`RETAIN_SAFETY_NULL`
E	1 $FS = 0$, 11 $FS = 1$	$p_{(1)} = 0.002122, p_{(2..12)} = 0.015523$	Step 2 ($0.015523 > 0.004652$)	1 / 12	Pass	`CONFIRMATORY_PASS_IUT`

4 Regression Verification and Conclusion

All 29 unit tests in `tests/test_v9_holdout_preflight.py` and all 140 regression tests in the repository pass with 100% success.

AUDIT DETERMINATION: H5_CONSISTENCY_CONFIRMED_WITH_QUALIFICATIONS

Qualifications: (1) The sequential stopping bug in `evaluate_scenario_holdout` has been resolved; (2) research publications must clearly distinguish the global conjunctive IUT status from scenario-level FWER claims; (3) no protocol amendment is needed because the implementation matches the registered decision rule.